Aakriti Shah

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Research Interests

I am passionate about leveraging machine learning for social good and building systems that contribute to a **better society**. My research interests span **Natural Language Processing**, **LLM interpretability**, machine unlearning, **trustworthy Machine Learning**, and medical imaging.

Education

MS **University of Southern California**, Computer Science — Aug 2024 - **Expected** May 2026

- GPA: 3.66
- **Coursework:** Trustworthy Machine Learning, Natural Language Processing, Deep Learning and Its Applications, Algorithms, Innovation for Defense Applications, Text as Data (NLP and LLMs)

BA **Rollins College**, Computer Science & Business Management (*Honors*) — Sept 2020 – Dec 2023

- GPA: 3.72
- **Coursework:** Operating Systems, Algorithms, Software Engineering, Computer Architecture, Data Science, Discrete Math, Software Engineering

Honors & Awards

Dean's List, University of Southern California (multiple semesters) — 2024-2025
Magna Cum Laude, Rollins College — 2023
NSF REU Awardee, Pennsylvania State University — 2023
Archibald Granville Bush Award: Excellent Standard in Academic Pursuit, Rollins College — 2023
Honors Degree Program, Rollins College — 2020-2023
President's List, Rollins College (multiple semesters) — 2020–2023

Publications & Submissions

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. *"Deep-learning-based segmentation of glands and stroma on digitized H&E-stained prostate whole slide images"*, **SPIE Medical Imaging 2026** (oral presentation). — February 2026, **Accepted for Oral Presentation**

Shah, A. (presenting author), Bogale, Y., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R. *"Automated gleason pattern segmentation in prostate whole slide digitized pathology using foundation model feature embeddings"*, **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster). — May 2026, **Accepted for Poster Presentation**

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R., *"Automated segmentation of gland and stromal morphology on digitized H&E-stained prostate whole slide images"*, **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster). — May 2026, **Accepted for Poster Presentation**

Shah, A., & Le, T. (2025). *"The Limits of Obliviate: Evaluating Unlearning in LLMs via Stimulus-Knowledge Entanglement–Behavior Framework"* arXiv preprint arXiv:2510.25732 Submitted to **EACL 2026/ACL RR 2026** in October 2025. — November 2025, **Under Review**

Shah, A. (2023). *"Evaluating AI Sentiment Analysis."* Rollins College Honors Program Theses, No. 218. Available at: scholarship.rollins.edu/honors/218/ — December 2023, **Honors Thesis**

Research Experience

Research Assistant, ARKAI Lab - Dr. Thai Le (Computer Science, Indiana University) — May 2025 – Ongoing, Project, Paper

- Investigating **targeted unlearning** in LLMs through **cognitive science frameworks**, exploring parallels between machine unlearning and human memory retrieval.

- Applying theories from psychology to analyze **knowledge entanglement** in unlearned LLMs, examining how unlearning affects associated concepts and resistance to **adversarial extraction attacks**.
- Developing evaluation frameworks to assess unlearning **robustness** and inform the design of more **secure LLMs**.

Research Assistant, AIMed Lab - Dr. Rakesh Shiradkar (Computer Science, Indiana University) May 2025 – Ongoing
Project Abstract

- Conducting computational pathology research on prostate **cancer detection** through automated tissue compartment and Gleason pattern segmentation in H&E-stained whole slide images.
- Extracted patches from and annotated glands from H&E-stained whole slide images using *QuPath*.
- Developed **novel postprocessing pipeline** incorporating lumen reconstruction to improve segmentation accuracy of tissue compartments (stroma, lumen, epithelium), achieving average Dice score of 0.895.
- Implementing and evaluating **deep learning and vision foundation architectures** (UNet, nnUNet, UNI) for robust multi-class segmentation across diverse histopathological datasets.
- Extending research to Gleason pattern grading (patterns 3, 4, 5) to enable deep learning automated cancer aggressiveness assessment.

Research Project, NLP and LLMs (University of Southern California) Aug 2025 – Dec 2025
Paper

- Used multiple unlearning algorithms (DPO, refusal tuning, and prompting) to remove political information from different LLMs (Llama-3.1-8B & Mistral-7B) to see the effects.
- These effects were measured using political compass and social psychology questionnaires (Pew Research Center Political Typology Test, Right-Wing & Left-Wing Authoritarianism, Social Dominance Orientation, and Openness to Experience in the Big Five).
- Specifically, if we remove information of politicians of a particular political party, does this shift the LLMs observed “political orientation” away from that ideology?

Research Assistant, Dr. Anita Penkova (Computer Science, University of Southern California) Sept 2024 – Dec 2025

- Conducting **computer vision** research for early and automated detection of Diabetic Retinopathy in retinal fundus images.
- Developing and optimizing preprocessing pipelines (CLAHE, Gaussian Blur, MGA-CSG) to address limited high-quality training data and **improve model generalization** across diverse image quality conditions.

Research Project, Natural Language Processing (University of Southern California) Jan 2025 – May 2025
Project Paper

- Developed Hinglish **conversational chatbot** by curating multilingual datasets with code-switched and synthetically generated training samples to address low-resource language challenges.
- **Fine-tuned instruction-tuned LLMs** for open-domain dialogue generation; evaluated outputs using syntactic and semantic metrics to assess code-switching quality and conversational coherence.

Honors Thesis, Dr. Daniel Myers (Rollins College) Jan 2023 – Dec 2023
Project Paper

- **Evaluated LLM performance** on sentiment analysis using manually labeled real-world interview data from non-profit Crave, comparing human-coded emotional assessments to model-generated outputs.
- Investigated **limitations of LLMs in capturing nuanced emotional context** in low-resource, qualitative settings, providing insights for responsible deployment in social science research.

Research Intern, National Science Foundation REU June 2023 – Aug 2023
Paper

- Investigated **adversarial robustness of ML models** for autonomous driving systems, analyzing vulnerabilities under perturbation attacks and evaluating transfer learning across architectures.
- Conducted interdisciplinary research, covering qualitative and quantitative methods, research ethics, and perspectives from computer science and psychology.

Technical Projects

Dog Rescue Website — Oct 2025 – Jan 2026, Website

- Developed a website for a dog rescue that runs completely on donations. Their only online presence previously was their Facebook page, and their growing rescue needed a bigger space to spread information about their adoptable dogs.

voteSmarter, Civic Engagement Mobile App — Feb 2024 – June 2024, Project

- Developed mobile application connecting users to political candidates based on values alignment and location; integrated features for voter registration, polling locations, and candidate databases.

ASLearning, ML for Accessibility — Jan 2023 – May 2023, Project

- Built mobile application using real-time object detection (TensorFlow) to facilitate communication between hearing-impaired and English-speaking users through sign language recognition.

Technical Skills

Programming: Python, SQL, Java, JavaScript, R, Assembly (ARM), C, C++, XML, Dart, C#, LaTeX, Swift
ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy
Tools & Platforms: Git, Docker, Jupyter, Google Cloud Platform, MongoDB, Flask, React, Node.js
Research Skills: Experiment design, data annotation, literature review, statistical analysis, scientific writing

Teaching and Outreach

Volunteer, Amy's Angels Dog Rescue — Sept 2025 - Ongoing, Website

- Volunteered to create a website for this dog rescue whose only online presence was Facebook.
- Needed a way to communicate their adoptable dogs and donation needs to their community.

Volunteer, Viterbi Impact — Aug 2024 – Ongoing

- Engaging in community service projects including fire clean-up, premature baby care, and food distribution.
- Helped distribute food for people who were affected by the 2025 Southern California Wildfires and the nationwide EDT crisis.

Volunteer, AAPI Coming Together (ACT) — May 2024 – Ongoing

- Supporting civic outreach and cultural advocacy within the Asian American community.
- Developed **voteSmarter** to support voter education efforts and civic engagement.

Instructor, iDTech Camps — June 2022 – Aug 2022

- Taught Java, Python, C#, and game development to K-12 students, promoting computational thinking and creative problem-solving in early learners.